HS1010: Foundations of Health & Safety and the Global Regulatory Landscape

🔷 LESSON SNAPSHOT

This 60-minute lesson introduces the **core foundations of Health & Safety (H&S)**, the **global regulatory landscape**, and the **ethical, legal, and management principles** that guide safe practice worldwide.

It integrates Ghana's legal framework and international bodies such as **NEBOSH, ILO, OSHA, and ISO** (particularly **ISO 45001**), and ends with a real case study — the **Rana Plaza disaster** — with reflection and guided discussion.

🎯 Learning Outcomes (after 60 minutes)

- ▶ Define "Health & Safety" and explain the difference between *hazard* and *risk*.
- ▶ Identify the roles and relationships of NEBOSH, ILO, OSHA, and ISO.
- ▶ Summarize the PDCA (Plan–Do–Check–Act) model of ISO 45001.
- ▶ Trace key historical milestones in the evolution of workplace safety.
- ▶ Recognize personal wellbeing actions and three essential HSE professional skills.

🔷 60-Minute Delivery Plan

Time	Segment	Key Focus
00:00 – 05:00	Welcome & Objectives	Introductions, student hazard examples
05:00 – 20:00	Core Concepts	Hazard vs Risk, Hierarchy of Control, Duty of Care
20:00 – 35:00	Global Frameworks	NEBOSH, ILO, OSHA, ISO — roles & interplay
35:00 – 45:00	ISO 45001 PDCA	Application to real workplaces
45:00 – 53:00	Case Study	Rana Plaza collapse + reflection
53:00 – 58:00	Discussion Quiz	5 key questions
58:00 – 60:00	Wrap-up	Summary, forum prompt, next steps

🔷 HS1011. Core Concepts & Definitions

ℹ️ **Health & Safety (H&S)** — The organized efforts, systems, and practices designed to prevent injury, illness, and promote the overall wellbeing of individuals in workplaces and public spaces.

Key Terms

▶️ **Hazard:** Any potential source of harm (e.g., chemicals, noise, wet floors).

▶️ **Risk:** The likelihood and consequence of harm resulting from a hazard.

▶️ **Control Measures:** Steps to eliminate or reduce risk.

▶️ **Hierarchy of Control:** Eliminate → Substitute → Engineer → Administrative → PPE.

▶️ **Duty of Care:** The legal and ethical obligation to protect others from harm.

▶️ **Wellbeing:** Encompasses mental health, fatigue, ergonomics, and lifestyle management.

⚠️ **Ghana Context (Callout Box)**

The **Labour Act 2003 (Act 651)** and **Factories, Offices & Shops Act 328** require employers to ensure safe systems of work, provide protective equipment, and report serious incidents.

Tip: Encourage practical, low-cost control measures and active reporting in informal sector settings.

🔷 HS1012. Global Health & Safety Frameworks

1️⃣ NEBOSH — The Knowledge Framework

▶️ Provides professional and vocational qualifications in occupational safety and health.

▶️ Aligns global HSE competency standards and helps employers identify skilled professionals.

2️⃣ ILO — The Policy and Labour Framework

▶️ Sets international conventions and recommendations (e.g., ILO-OSH 2001).

▶️ Guides national policies and social dialogue for decent, safe work.

3️⃣ OSHA — The Enforcement Model (USA)

▶️ Offers practical training materials and inspection guidance.

▶️ Inspires national authorities globally, including Ghana's Department of Factories Inspectorate.

4️⃣ ISO — The Management-System Framework

▶️ **ISO 45001** provides a systematic, auditable structure for occupational safety management.

▶️ Encourages leadership, worker participation, and continuous improvement.

How They Interact

ILO sets global standards → ISO 45001 operationalizes them in management systems → NEBOSH professionalizes the workforce → national regulators (like OSHA or Ghana DFI) enforce compliance.

🔷 HS1013. Decoding ISO 45001 — The PDCA Model

PLAN — Establish context, identify hazards, set measurable objectives, prepare risk register.

DO — Implement training, communication, operational controls, documentation.

CHECK — Monitor performance, conduct audits, review incidents.

ACT — Implement corrective actions and management reviews for continual improvement.

☑ *Practical Templates (to attach in LearnDash)*

1. Simple Risk-Register (columns: hazard, consequence, likelihood, control, owner, review date)
2. Training Matrix (who needs what and when)
3. 5-Item Audit Checklist (quick internal review tool)

🔷 HS1014. The Evolution & Future of Health & Safety

▶ **Pre-1900s:** Industrial revolution hazards, child labour, early welfare reforms.

▶ **1900s–1950s:** Factory Acts, occupational medicine.

▶ **1970s–1990s:** Rise of OSHA, ILO conventions, international cooperation.

▶ **2000s–Present:** ISO 45001 systems thinking, mental health focus, ESG reporting.

▶ **Future:** AI-based risk prediction, remote-work ergonomics, global supply-chain accountability.

🔷 HS1015. Case Study — Rana Plaza (Bangladesh, 2013)

⚠️ **Summary:**

An eight-storey garment factory collapsed due to ignored structural warnings, killing over 1,100 workers. It remains one of the deadliest industrial disasters in history.

Key Lessons for Practitioners

▶ Safety culture must be reinforced by engineering integrity and accountability.

▶ Supply-chain oversight and brand responsibility save lives.

▶ Worker participation and inspection systems are vital for prevention.

Verified Case Study Video

"*Rana Plaza: The Collapse That Shook the World*" — DW Documentary

https://www.youtube.com/watch?v=Mq_mTnhoxEY (runtime ≈ 12 min 23 sec)

Class Activity (45 – 53 min)

– Watch 7–8 minutes of the video (teacher cue).

– Discuss in breakout forum:

- Three immediate preventive actions.
- Two long-term system improvements (policy or training).

Reflection Prompt (Forum):

If ISO 45001 had been applied, which clauses (leadership, worker participation, hazard identification, procurement) could have prevented the disaster?

🔷 HS1016. Personal Wellbeing & Professional HSE Skills

Checklist for Everyday Safety Leadership

- ▶ Maintain healthy routines — sleep, nutrition, hydration, mental rest.
- ▶ Use active observation — identify and report hazards early.
- ▶ Communicate clearly — safety depends on shared understanding.
- ▶ Investigate ethically — focus on causes, not blame.
- ▶ Practice empathy and cultural awareness.

Mini-Activity (Forum):

Keep a one-week "Wellbeing & Safety Log." Each day, note one personal wellbeing action and one workplace or community hazard observation.

🔷 HS1017 - Discussion Quiz & Resources (5 Questions)

1. Define "hazard" and "risk," giving one Ghanaian example.
2. List the hierarchy of control in order and provide one engineering control example.
3. Name two functions of NEBOSH and one Ghana regulation protecting workers.
4. From the Rana Plaza case, identify two root causes and two preventive measures.
5. Which ISO 45001 clause would you prioritise in a small manufacturing firm, and why?

(5-minute group or forum discussion; brief answers posted to LearnDash forum.)

🎥 Recommended Verified Video Resources (Checked 2025-11-05)

▶ NEBOSH Overview (Official Channel)

"*What is NEBOSH | Benefits of NEBOSH*" —

https://www.youtube.com/watch?v=aqezK6LgWRE (≈ 5 min 30 sec)

▶ ILO OSH Management Systems (Official ILO)

"*Occupational health and safety management system || ILO OSH 2001*" —

https://www.youtube.com/watch?v=7Rg0cOBS-RE (≈ 13 min 36 sec)

Both are educational, publicly available, and suitable for embedding within LearnDash.

🔷 References & Verified Sources (Checked 2025-11-05)

1 **DW Documentary:** *"Rana Plaza: The Collapse That Shook the World"* — https://www.youtube.com/watch?v=Mq_mTnhoxEY (12 min 23 sec).
2 **ILO:** *"Occupational health and safety management system || ILO OSH 2001"* — https://www.youtube.com/watch?v=7Rg0cOBS-RE (13 min 36 sec).
3 **NEBOSH Official Channel:** https://www.youtube.com/@neboshofficial — sample intro https://www.youtube.com/watch?v=aqezK6LgWRE (5 min 30 sec).
4 **ISO 45001:** Overview — https://www.iso.org/iso-45001-occupational-health-and-safety.html
5 **NEBOSH Resources:** https://www.nebosh.org.uk/our-news-and-events/our-news/webinars-and-videos/
6 **OSHA Resources:** https://www.osha.gov/video/
7 **Ghana Legislation:** *Labour Act 2003 (Act 651)*; *Factories, Offices & Shops Act 328.*

ℹ *All multimedia links verified as publicly available and embeddable as of 5 Nov 2025. Instructors should preview videos before each class to cue relevant segments.*
